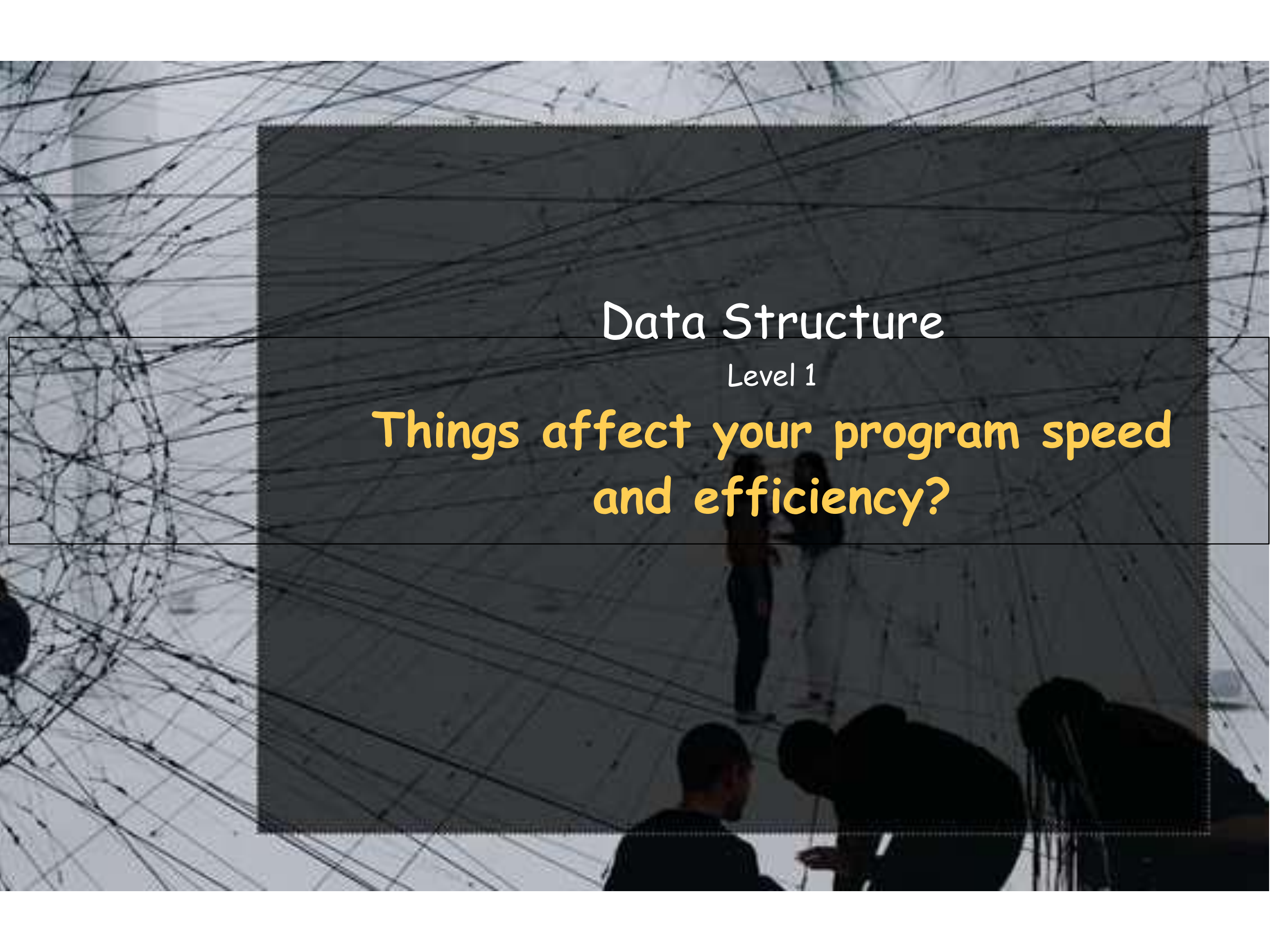


Data Structure

Level 1

Time & Space Complexity
Big O Notation

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing a light blue-grey field. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. A large, dark, semi-transparent rectangular box is centered on the slide, serving as a backdrop for the text.

Data Structure

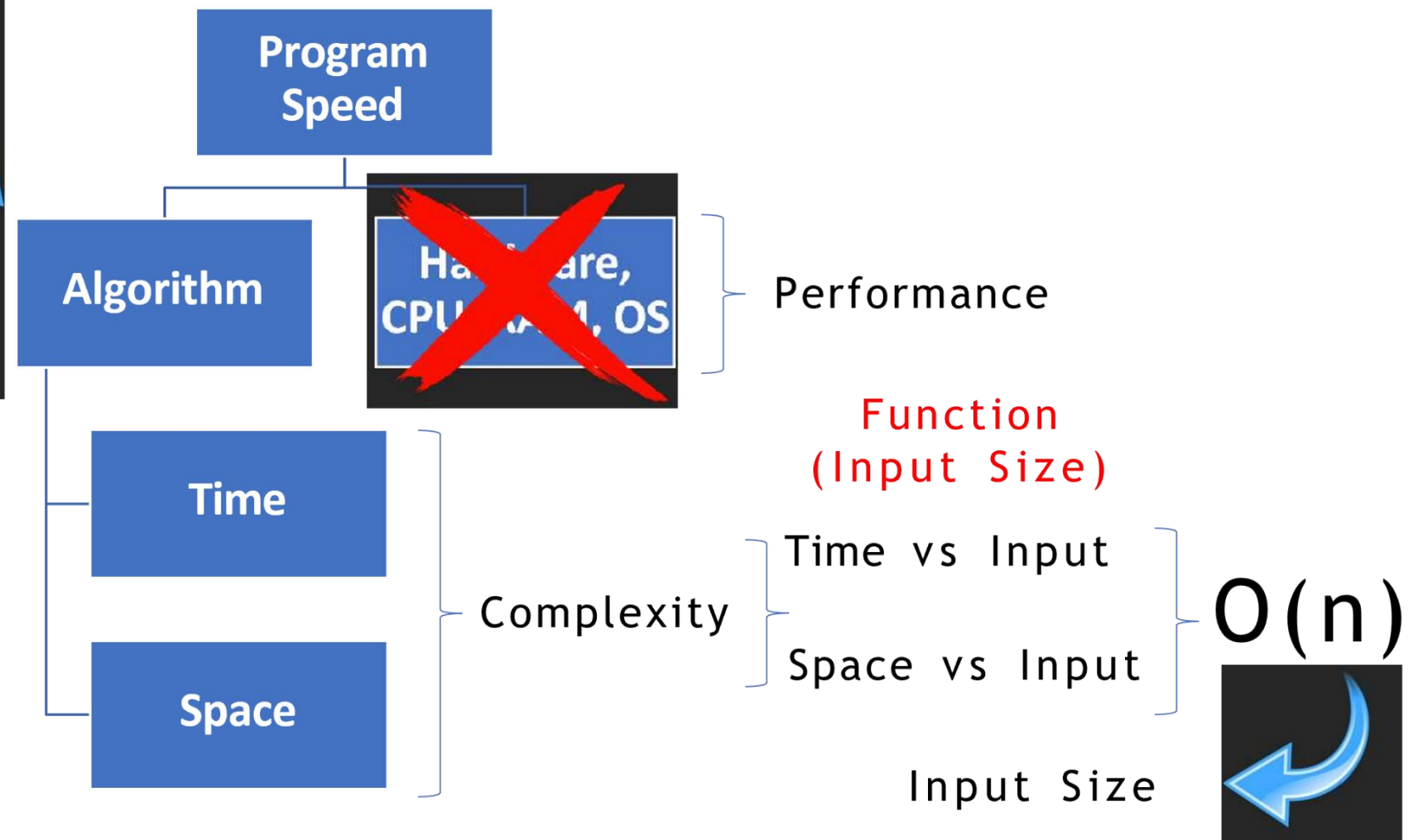
Level 1

Things affect your program speed
and efficiency?

Things Affect Program Speed



Printing Array[10] Items
Is faster than
Printing Array[1000] Items



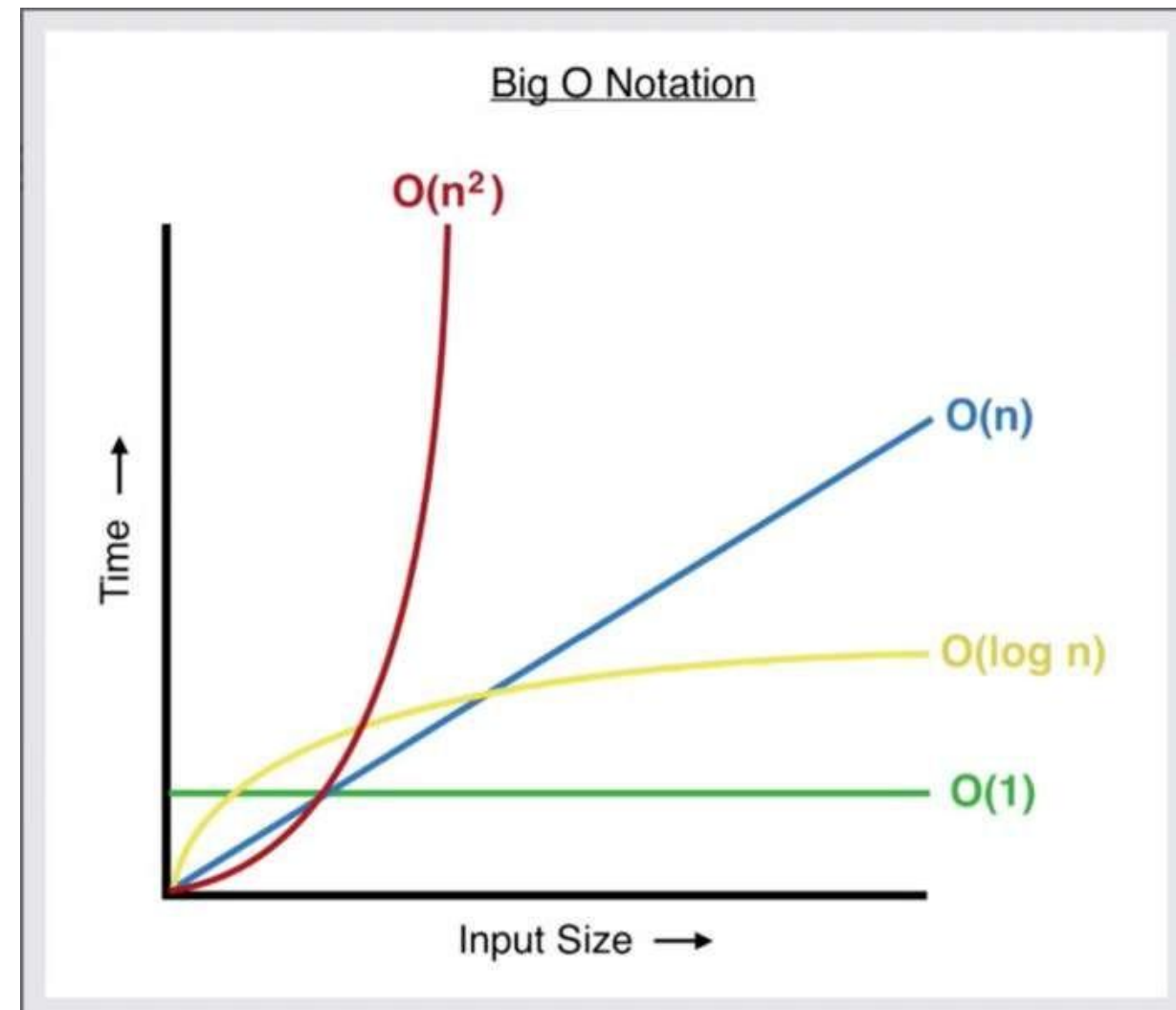
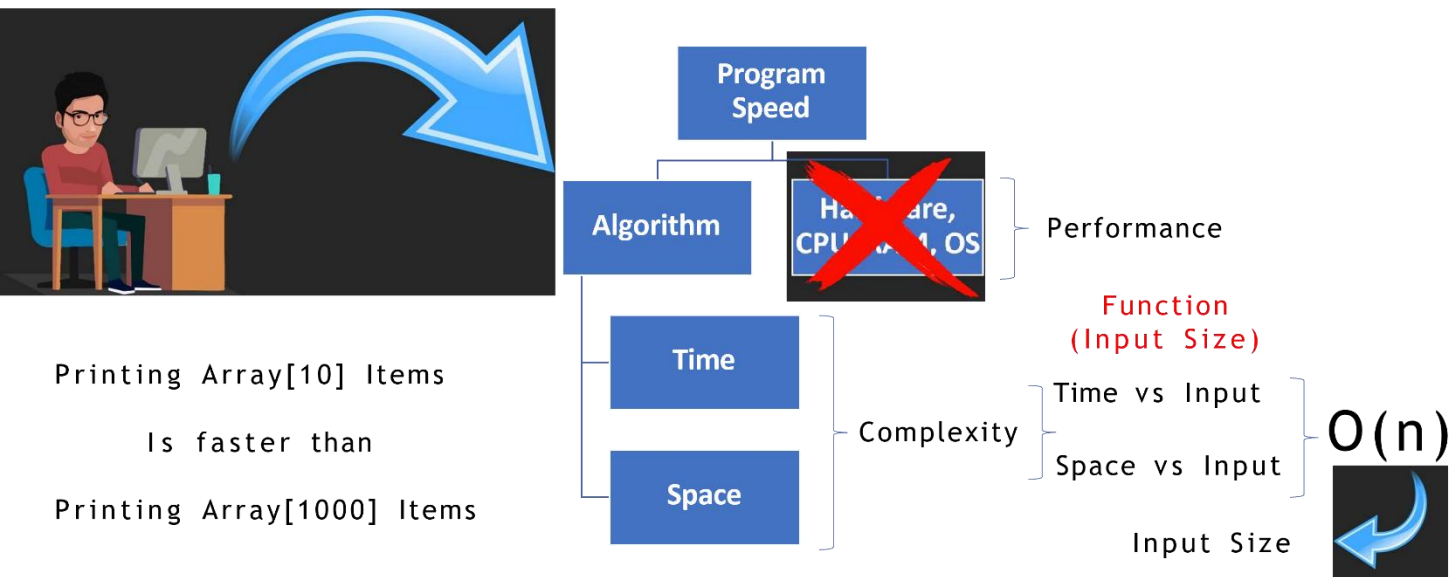
The background of the slide features a complex network of thin, dark lines crisscrossing over a lighter, textured surface, resembling a web or a circuit board. In the lower portion of the image, the dark silhouettes of several people are visible, appearing to be in a collaborative setting, possibly a meeting or a workshop, with one person standing and others seated or leaning over. A large, dark, semi-transparent rectangular box is centered on the slide, serving as a backdrop for the text.

Data Structure

Level 1

Big O

What is Big O?



The background of the slide features a complex, abstract network of thin, dark lines crisscrossing a light gray background. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. A large, dark gray rectangular area is centered on the slide, serving as a backdrop for the text.

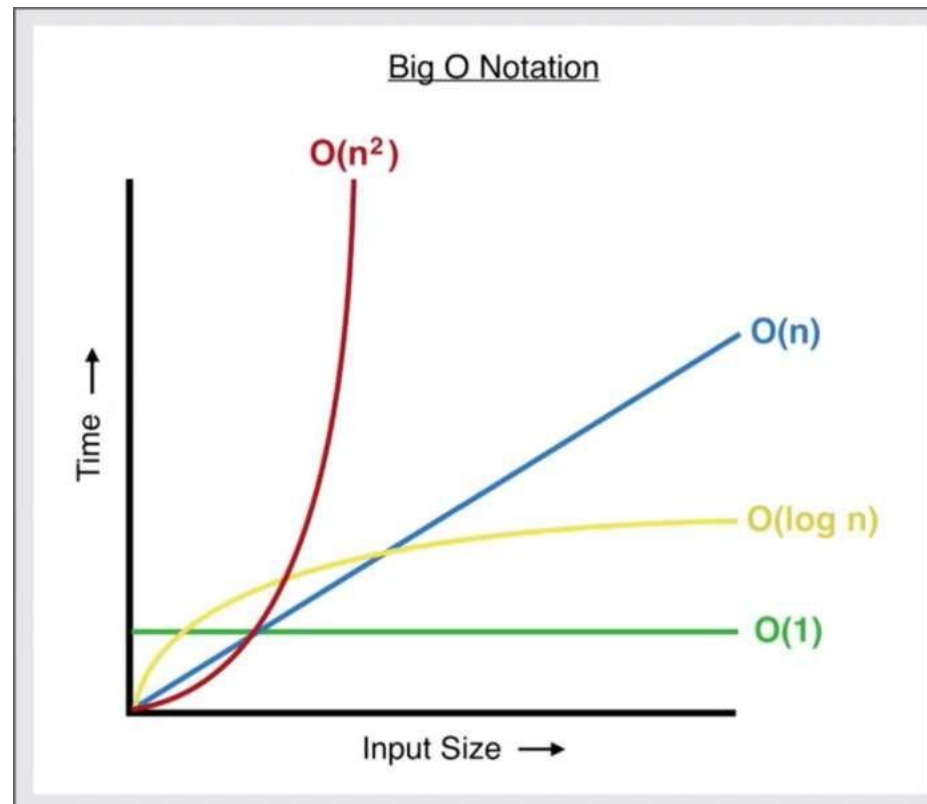
Data Structure

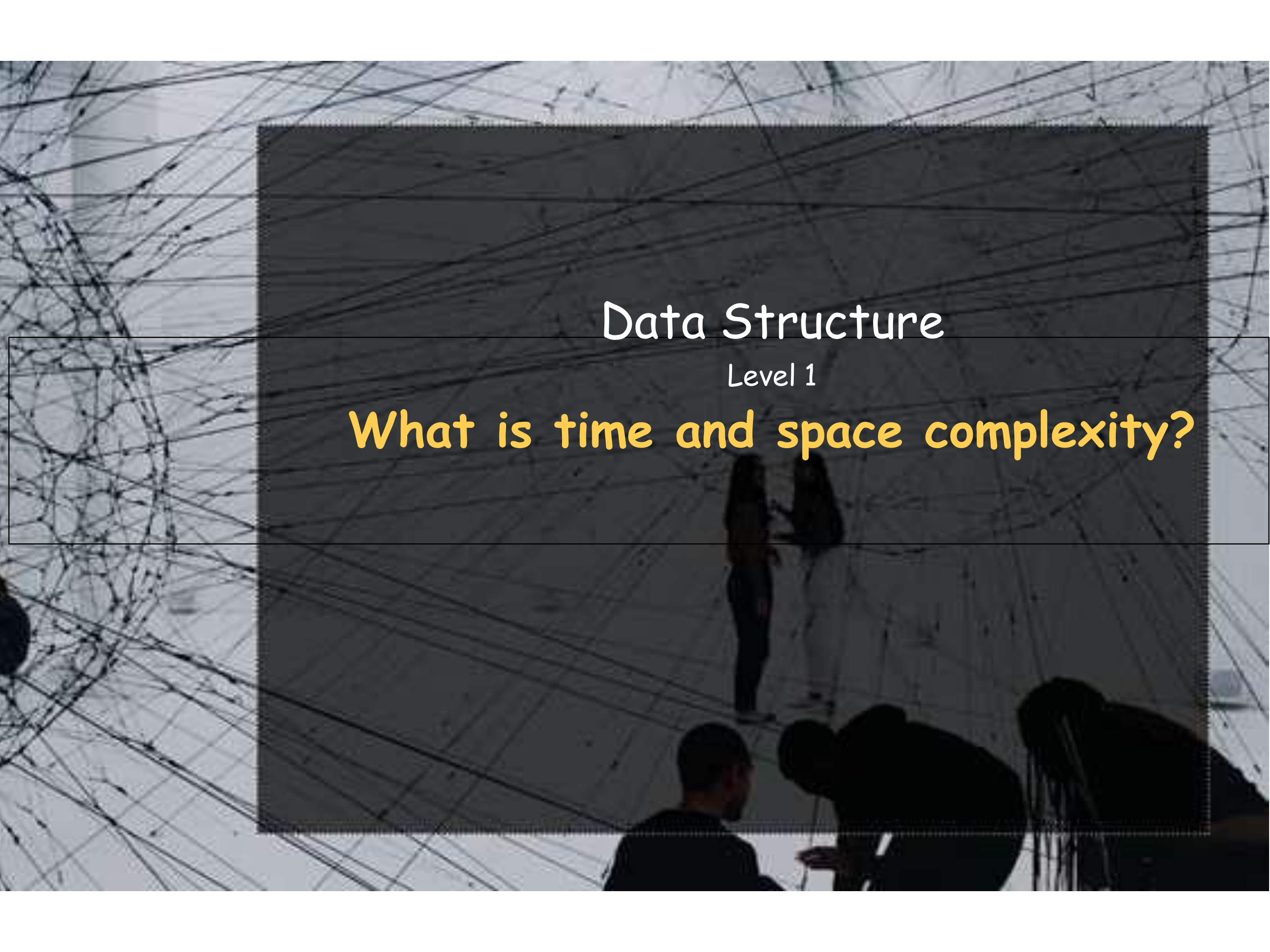
Level 1

What Big O is NOT?

What is Big O is NOT?

- Big O is not going to give you an exact answer on how long a piece of code will take to run.
- Does not consider the performance of hardware.



The background of the slide features a complex, abstract network of thin, dark lines crisscrossing a light gray background, resembling a web or a data structure. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. A large, dark gray rectangular area is overlaid on the right side of the image, serving as a backdrop for the text.

Data Structure

Level 1

What is time and space complexity?

Time & Space Complexity

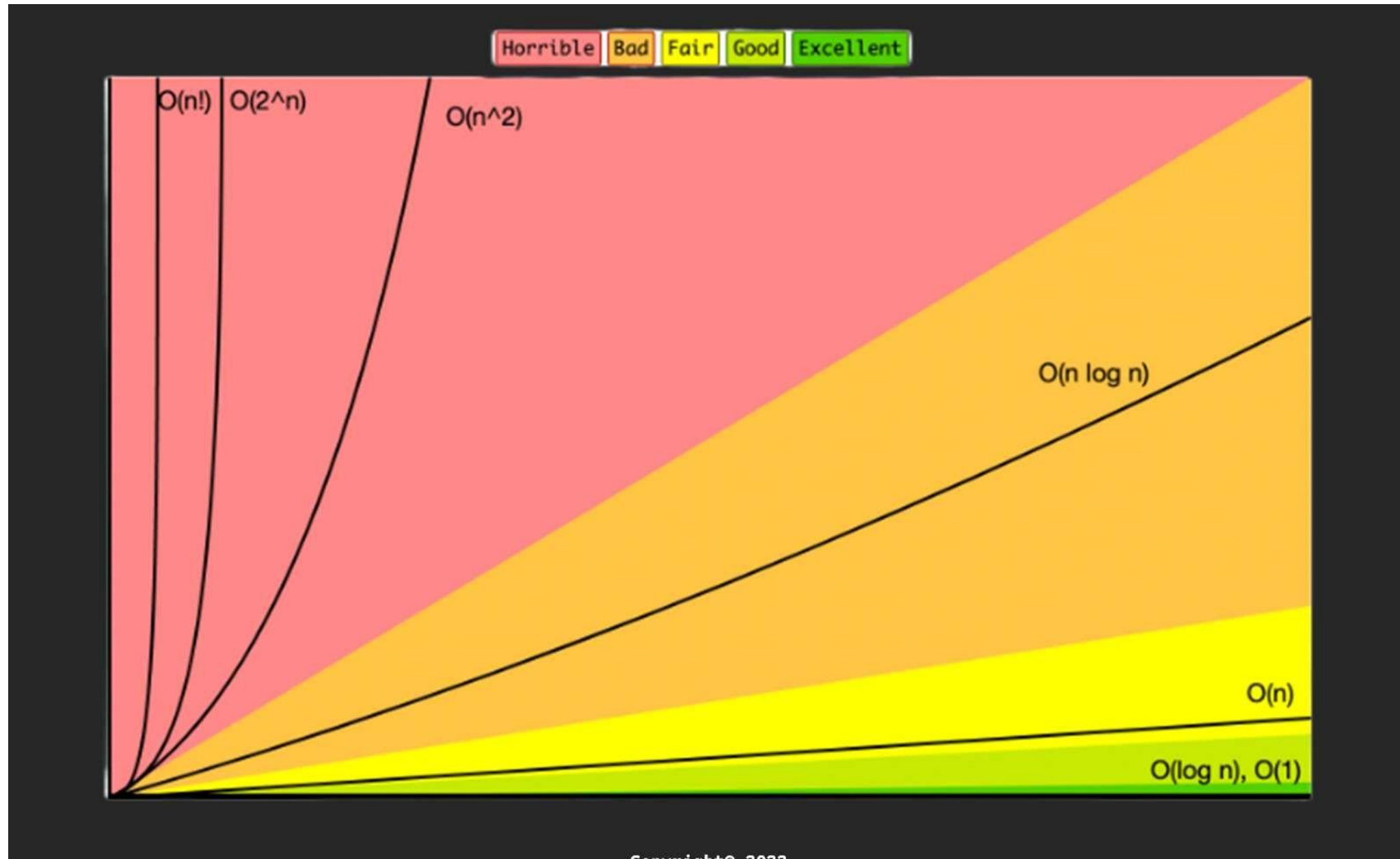
An algorithm's time complexity: specifies how long it will take to execute an algorithm as a function of its input size.

Similarly, an algorithm's space complexity: specifies the total amount of space or memory required to execute an algorithm as a function of the size of the input.

What is BigO Notation

- The big-O originally stands for "Order Of".
- Efficiency of Algorithm
- Time Factor of Algorithm
- Space Complexity of Algorithm
- Represented by $O(n)$ where n is the number of Inputs.
- Big O, also known as Big O notation, represents an algorithm's worst-case complexity.
- It uses algebraic terms to describe the complexity of an algorithm.
- Big O defines the runtime required to execute an algorithm, by identifying
 - how the performance of your algorithm will change as the input size grows.
- Relationship between Input Size and Performance.
- But it does not tell you the exact time your algorithm's runtime is.
- Big O notation measures the efficiency and performance of your algorithm using time and space complexity.
- Big O allows us to discuss our code algebraically to get a sense of how quickly it might operate under the strain of large data sets.
- with Big O notation, we can look at our algorithm and see that it will take $O(n)$ time to run. Big O notation, written as $O(\text{blank})$, show us how many operations our code will run, and how its runtime grows in comparison to other possible solutions.

What is BigO



Thank you 😊